

Traffic Rule Compliance Engine

Behaviour-science design to shift youth driving norms — beyond fines



Problem Snapshot

Root causes

Social norms, peer signalling,
low perceived risk

Why fines fall short

Low detection, punished
individuals not groups,
resilience of norms

Opportunity

Leverage incentives, social accountability, micro-nudges



Design Principles



Gain-framed incentives

Positive rewards for repeated safe behaviour



Peer accountability

Small-group reputation scores — social recognition



Contextual nudges

Real-time prompts at risk moments (speed, phone use)



Measure & iterate

Frequent, transparent metrics to refine interventions





System Overview — The Engine

Three integrated layers: detect → nudge → socialise & reward

01

Detect

Mobile telematics + roadside sensors + self-reports

02

Nudge

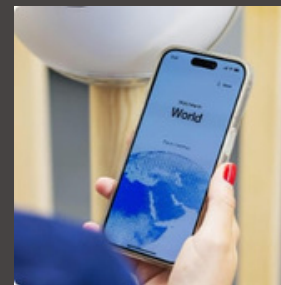
Real-time contextual messages, haptic feedback, delayed summaries

03

Socialise & Reward

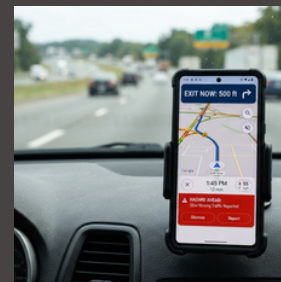
Small-group leaderboards, public badges, redeemable micro-rewards

Core Intervention Components



Daily Safe Streaks

Small tokens earned for each compliant trip



Smart Nudges

Time/place prompts and gentle haptics at risk points



Peer Pods

3–8 friends share a weekly reputation score

Incentive Design — Behavioural Levers



Immediate micro-rewards

Points convertible to fuel discounts, phone data



Social recognition

Public badges, local NGO endorsements



Milestone bonuses

Monthly cashbacks or insurance premium rebates





Metrics — What We Track

Behavioural KPIs

Helmet use %, phone-use while driving %, overspeed events per 1,000 km

Engagement KPIs

Daily active pods, streak retention (7/30/90 day), reward redemptions

Social KPIs

Peer reports frequency, badge-sharing rate, community nominations

Outcome KPIs

Reduction in crashes per 10k trips, hospital admissions, sustained change

Pilot Roadmap



Safeguards & Equity



Privacy-first

Local data residency, opt-in telematics, minimal location retention

Equitable rewards

Non-monetary recognition and in-kind benefits for low-income youth

Non-punitive

Use learning feedback not automated fines; partner enforcement only if needed



Next Steps & Calls to Action



1. Convene stakeholders

City, NGOs, insurers, telco partners



2. Launch pilot

50–200 youth, 3 months, pre/post evaluation



3. Evaluate & scale

Use KPIs to optimise, scale city → state



Goal: Make safe driving socially desirable — measurable reductions in risky behaviours within 6 months.